SIMATS Engineering

SIMATS Online Programming Lab — Python Programming

Course Name	Python Programming
Course Code	CSA0804
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Register Number	1972260035B
Topic	CO5-AT2-DEBUGGING AND OPTIMIZATION TASK
Question	Q4 — Matplotlib Plot Crash (15 Marks) — Matplotlib – Plotting, Labels, Save

Q4. Matplotlib Plot Crash [5 Marks] — Matplotlib – Plotting, Labels, Save

■ This program contains 5 bug(s). Identify, fix, and optimise.

(a) Identify errors that prevent the plot from rendering correctly.

(b) Fix and produce a properly labelled bar chart with title, x-label, y-label, and grid.

(c) Save the figure as 'sales_chart.png' at 300 DPI.

Buggy Program:

```
import matplotlib.pyplot as plt
import pandas as pd
df = pd.read_csv('monthly_sales.csv')
x = df['Month']
y = df['Sales']
plt.bar(y, x, color='steelblue')
plt.title('Monthly Sales')
plt.xlabel = 'Month'
plt.ylabel = 'Sales (Rs)'
plt.xticks(rotation=90, fontsize=8)
plt.tight_layout
plt.savefig('sales_chart', dpi=300)
plt.show
```

(a) Errors That Prevent the Plot From Rendering Correctly

Bug 1 – bar() arguments swapped (Line 6):

plt.bar(y, x, color='steelblue') passes **Sales** as the x-positions and **Month** as the bar heights — the reverse of what's intended. Since **Month** is text, using it as a numeric bar height raises an error (or plots nonsense if coerced). It should be **plt.bar(x, y, color='steelblue')**.

Bug 2 – xlabel assigned instead of called (Line 8):

plt.xlabel = 'Month' overwrites the **xlabel** function itself with a plain string instead of invoking it, so the x-axis label is never actually set. It should be **plt.xlabel('Month')**.

Bug 3 – ylabel assigned instead of called (Line 9):

Same issue as Bug 2: **plt.ylabel = 'Sales (Rs)'** replaces the function with a string rather than calling it. It should be **plt.ylabel('Sales (Rs)')**.

Bug 4 – Missing parentheses on tight_layout() and show():

plt.tight_layout and **plt.show** both reference the function objects without calling them, so the layout is never adjusted and the figure window is never opened. Both need parentheses: **plt.tight_layout()** and **plt.show()**.

Bug 5 – savefig() missing file extension (part c):

plt.savefig('sales_chart', dpi=300) is missing the **.png** extension the task requires. It should be **plt.savefig('sales_chart.png', dpi=300)**.

(b) Corrected and Optimised Program

The program below fixes all five bugs, produces a properly labelled bar chart with title, x-label, y-label, and a grid, and saves the figure as required in part (c).

```
"""
CO5-AT2 - Debugging and Optimisation Task
Challenge 4: Matplotlib Plot Crash [Matplotlib - Plotting, Labels, Save]
------------------------------------------------------------
Loads monthly sales data from CSV and renders a labelled bar
chart of Sales by Month, with a grid, then saves the figure
as sales_chart.png at 300 DPI.
"""

import matplotlib.pyplot as plt
import pandas as pd

# ---------------------------------------------------------------
# 1. Load the CSV file
# ---------------------------------------------------------------
df = pd.read_csv('monthly_sales.csv')
x = df['Month']
y = df['Sales']

# ---------------------------------------------------------------
# 2. Draw the bar chart (fixed: x and y in the correct order)
# ---------------------------------------------------------------
plt.bar(x, y, color='steelblue')

# ---------------------------------------------------------------
# 3. Labels, title, and grid
#    (fixed: xlabel()/ylabel() called, not assigned)
# ---------------------------------------------------------------
plt.title('Monthly Sales')
plt.xlabel('Month')
plt.ylabel('Sales (Rs)')
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.xticks(rotation=90, fontsize=8)

# ---------------------------------------------------------------
# 4. Layout, save, and display
#    (fixed: tight_layout()/savefig()/show() all called properly,
#    with the .png extension required by part (c))
# ---------------------------------------------------------------
```

```
plt.tight_layout()
plt.savefig('sales_chart.png', dpi=300)
plt.show()
```

(c) Saving the Figure

plt.tight_layout() is called (with parentheses) before saving so the rotated month labels are not clipped.

plt.savefig('sales_chart.png', dpi=300) writes the figure to disk as a PNG at 300 DPI, exactly as required.

plt.show() is called last so the figure is both saved and displayed.

plt.grid(axis='y', linestyle='--', alpha=0.5) adds the gridlines requested in part (b), on top of the title and both axis labels.

Expected Result

Exact bar heights depend on the contents of **monthly_sales.csv**. For a file with 12 months of *Month, Sales* data, running the corrected program produces a vertical bar chart with months along the x-axis (rotated 90°), sales in rupees on the y-axis, a dashed horizontal grid, the title “Monthly Sales”, and no errors or warnings in the console:

```
(No console errors - the chart window opens showing 12 bars,
one per month, with the y-axis labelled "Sales (Rs)" and the
x-axis labelled "Month".)

File written: sales_chart.png (300 DPI)
```

Conclusion

The original script had its bar() arguments reversed, replaced both axis-label functions with plain string assignments, left **tight_layout()** and **show()** uncalled, and saved the figure without a file extension. None of these raise a single obvious traceback together, which is what made the plot silently fail to render as intended. The corrected version restores the right argument order in **bar()**, calls every pyplot function it uses, adds a grid for readability, and saves the chart as **sales_chart.png** at 300 DPI as required.